3. Tertiary structure:

The secondary structure elements (α-helix & β-sheets) further folds to form the three dimensional arrangement. This structure is called tertiary structure of the polypeptide (protein). Tertiary structure of proteins are stabilised by the interactions between the side chains of the amino acids. These interactions include the disulphide bridges between cysteine residues, electrostatic, hydrophobic, hydrogen bonds and van der Waals interactions.

4. Quaternary Structure

Some proteins are made up of more than one polypeptide chains. For example, the oxygen transporting protein, haemoglobin contains four polypeptide chains while DNA polymerase enzyme that make copies of DNA, has ten polypeptide chains. In these proteins the individual polypeptide chains (subunits) interacts with each other to form the multimeric structure which are known as quaternary structure. The interactions that stabilises the tertiary structures also stabilises the quaternary structures.

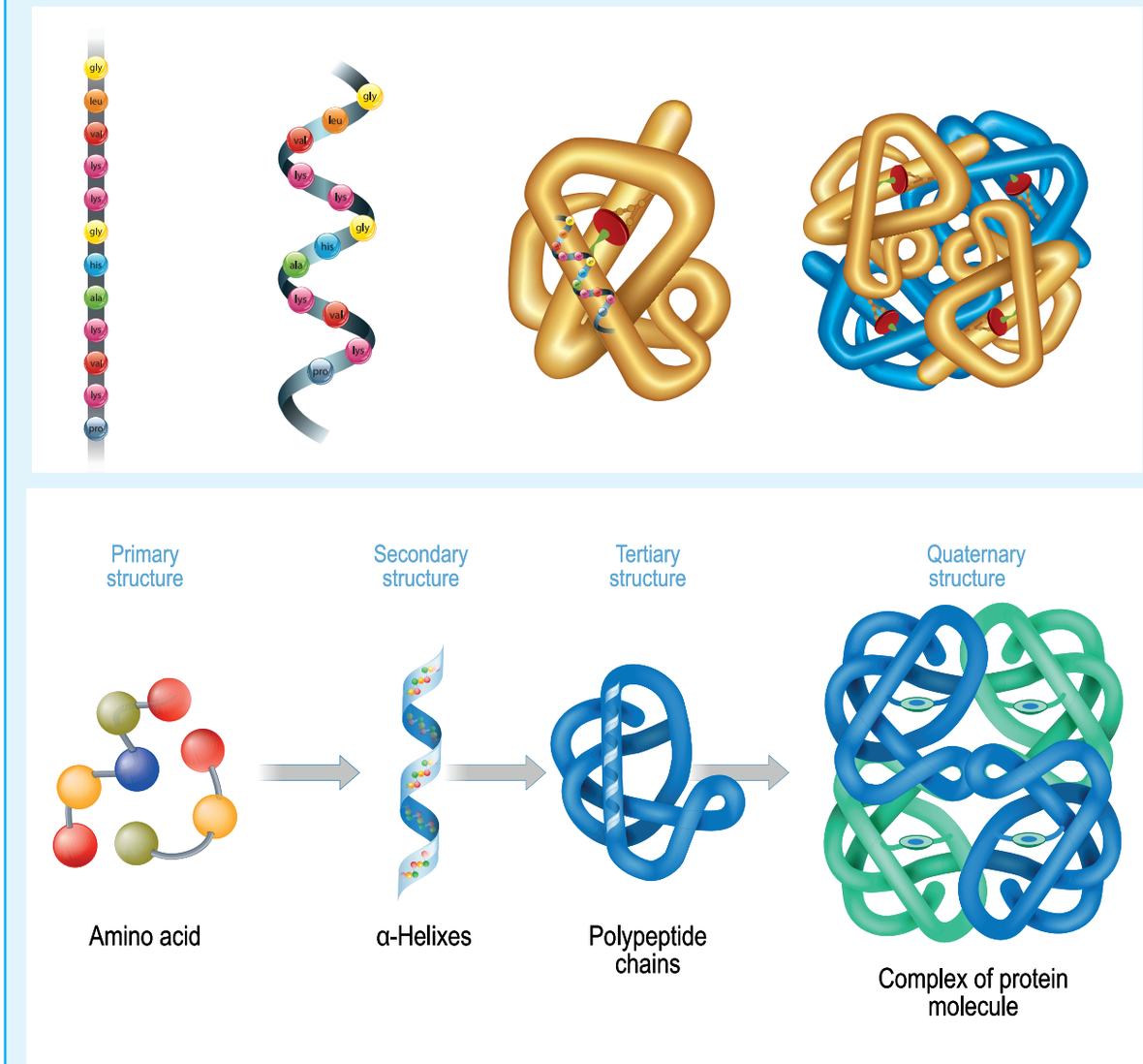


Figure 14.17 Four levels of protein structure

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